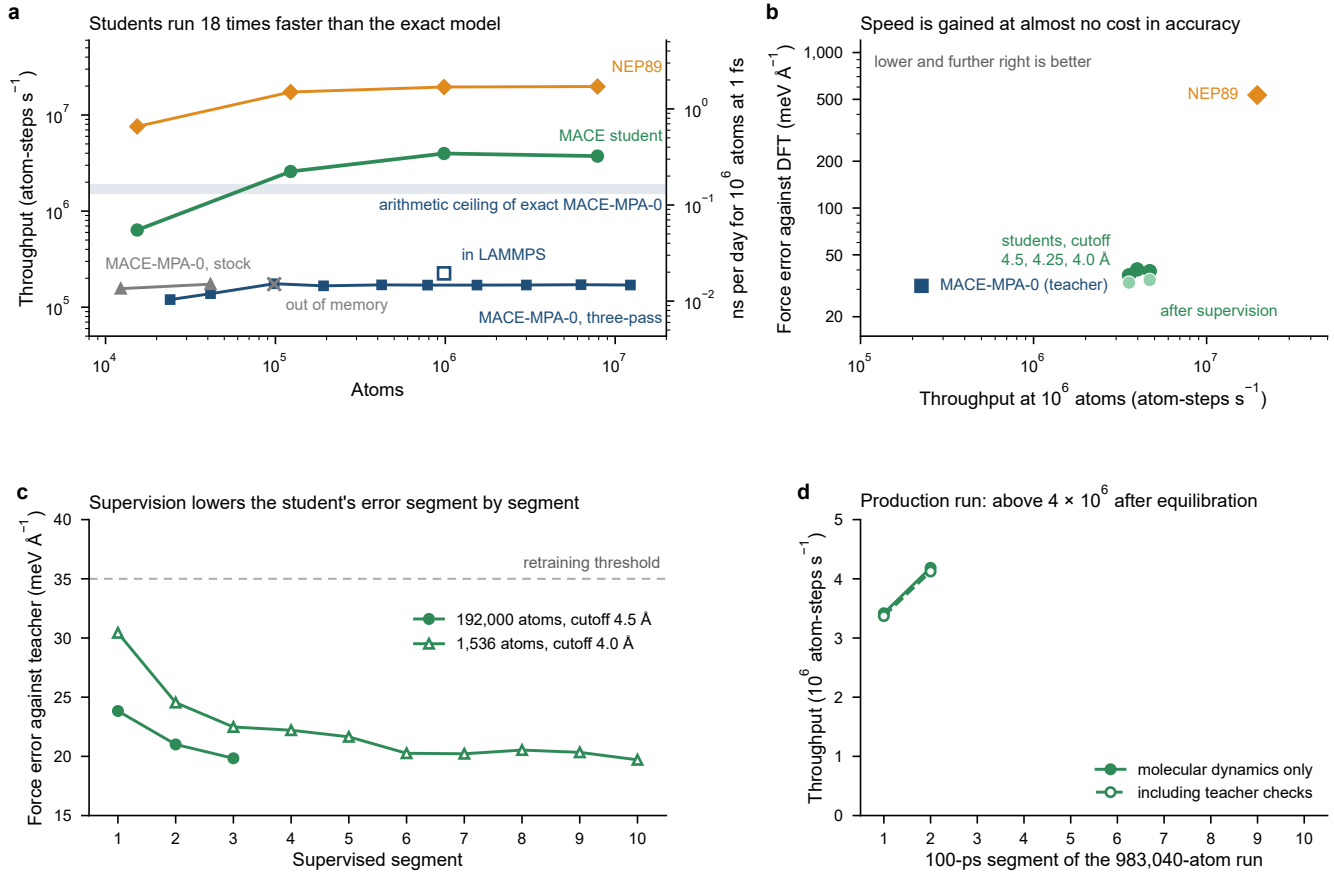
a
Students run 18 times faster than the exact model
NEP89
MACE student
arithmetic ceiling of exact MACE-MPA-0
in LAMMPS
MACE-MPA-0, stock
out of memory
MACE-MPA-0, three-pass
Throughput (atom-steps s$^{-1}$)
ns per day for $10^6$ atoms at 1 fs
Atoms
b
Speed is gained at almost no cost in accuracy
lower and further right is better
NEP89
students, cutoff 4.5, 4.25, 4.0 Å
MACE-MPA-0 (teacher)
after supervision
Force error against DFT (meV Å$^{-1}$)
Throughput at $10^6$ atoms (atom-steps s$^{-1}$)
c
Supervision lowers the student's error segment by segment
retraining threshold
192,000 atoms, cutoff 4.5 Å
1,536 atoms, cutoff 4.0 Å
Force error against teacher (meV Å$^{-1}$)
Supervised segment
d
Production run: above 4 × $10^6$ after equilibration
molecular dynamics only
including teacher checks
Throughput ($10^6$ atom-steps s$^{-1}$)
100-ps segment of the 983,040-atom run